

Xiangchi Yuan

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RESEARCH INTERESTS	Reasoning and Agentic LLM Post-Training (SFT and RL), Efficient Long-Context Modeling	
EDUCATION	Georgia Institute of Technology , Atlanta, GA Ph.D. in Computer Science Advisor: Professor Wenke Lee	August 2024–2027
	Brandeis University , Waltham, MA M.S. in Computer Science Brandeis University Merit Scholarship	May 2024
	University of Glasgow & UESTC , Glasgow, UK & Chengdu, CN B.Eng. in Electronic and Information Engineering	June 2022
EXPERIENCE	ByteDance Seed , San Jose, CA Research Intern Host: Dr. Ming Tu , Dr. Lu Lu Core contributor to the multimodal RFT/RL recipe for the audio understanding component of Seed (Doubao) flagship model , including data-mixture design, advantage aggregation, and process-reward optimization with PPO/VAPO; improved audio-text understanding by 5.1% on average across internal and public benchmarks (MMAU, MMSU), outperforming previous SOTA Gemini 3.1 Pro in this domain. Built an agentic audio framework (thinking-with-audio) for training and evaluation from scratch for long-form speaker diarization, reducing DER from 55.7% to 11.3% on the Seed Omni model.	May 2026–
	Adobe Research , San Jose, CA Research Intern Host: Dr. Xiang Chen Developed an efficient post-training framework for reasoning agents that dynamically interleaves RL and SFT while mitigating cross-stage forgetting, achieving state-of-the-art reasoning performance with only 7.8% of the training data. Published at EMNLP 2026 and filed a U.S. patent.	May–August 2025
PREPRINTS	Priority-Preserving Advantage Aggregation for Multi-Reward Policy Optimization [Code] <i>Preprint, 2026</i> Xiangchi Yuan et al. Introduces priority-preserving advantage aggregation for multi-reward RL, improving auxiliary objectives while preserving the optimization direction of the primary objective.	[PDF]
SELECTED PUBLICATIONS	Behavior Knowledge Merge in Reinforced Agentic Models <i>The 64th Annual Meeting of the Association for Computational Linguistics (ACL Main), 2026</i> Xiangchi Yuan , Dachuan Shi, Chunhui Zhang, Zheyuan Liu, Shenglong Yao, Soroush Vosoughi, Wenke Lee	[PDF] [Code]

Introduces the first model-merging framework for combining specialized RL-trained agents into a single generalist model that outperforms the original experts in their respective domains.

Mitigating Forgetting Between Supervised and Reinforcement Learning Yields Stronger Reasoners [PDF]

Conference on Empirical Methods in Natural Language Processing (EMNLP Main), 2026

Xiangchi Yuan, Xiang Chen, Tong Yu, Dachuan Shi, Can Jin, Wenke Lee, Saayan Mitra

Interleaves SFT and RL for reasoning post-training while protecting RL-acquired knowledge from catastrophic forgetting, achieving SOTA performance with only 1.5% of SFT data and 20.4% of RL.

SwiReasoning: Switch-Thinking in Latent and Explicit for Pareto-Superior Reasoning Large Language Models [PDF] [Code]

International Conference on Learning Representations (ICLR), 2026

Dachuan Shi, Abedelkadir Asi, Keying Li, **Xiangchi Yuan**, Leyan Pan, Wenke Lee, Wen Xiao

Combines latent and explicit reasoning through adaptive switch-thinking, improving reasoning accuracy by 3.1% while increasing average token efficiency by 79%.

Superficial Self-Improved Reasoners Benefit from Model Merging [PDF] [Code]

Conference on Empirical Methods in Natural Language Processing (EMNLP Main), 2025

Xiangchi Yuan, Chunhui Zhang, Zheyuan Liu, Dachuan Shi, Leyan Pan, Soroush Vosoughi, Wenke Lee

Identifies superficial self-improvement in reasoning LLMs and introduces iterative model merging with backfill RFT to improve reasoning generalization.

LaCache: Ladder-Shaped KV Caching for Efficient Long-Context Modeling of Large Language Models [PDF] [Code]

International Conference on Machine Learning (ICML), 2025

Dachuan Shi, Yonggan Fu, **Xiangchi Yuan**, Zhongzhi Yu, Haoran You, Sixu Li, Xin Dong, Jan Kautz, Pavlo Molchanov, Yingyan Celine Lin

Introduces a training-free ladder-shaped KV caching framework with iterative compaction for memory-efficient long-context LLM inference and generation.

Mitigating Emergent Robustness Degradation while Scaling Graph Learning [PDF] [Code]

International Conference on Learning Representations (ICLR), 2024

Xiangchi Yuan*, Chunhui Zhang*, Yijun Tian, Yanfang Ye, Chuxu Zhang

Develops a scalable MoE framework that mitigates emergent robustness degradation as adversarial attack strength increases.

Graph Cross-Supervised Learning via Generalized Knowledge [PDF] [Code]

ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD Oral), 2024

Xiangchi Yuan, Yijun Tian, Chunhui Zhang, Yanfang Ye, Nitesh V. Chawla, Chuxu Zhang

Introduces a cross-supervised MoE model that can leverage generalized knowledge for new data.

SELECTED
COLLABORATIONS

Steering Multimodal Large Language Models Decoding for Context-Aware Safety [PDF] [Code]

Conference on Empirical Methods in Natural Language Processing (EMNLP Main), 2026

Zheyuan Liu, Zhangchen Xu, Guangyao Dou, **Xiangchi Yuan**, Zhaoxuan Tan, Radha Poovendran, Meng Jiang

MetaState: Persistent Working Memory Enhances Reasoning in Discrete Diffusion Language Models [PDF] [Code]

Conference on Language Modeling (COLM), 2026

Kejing Xia, Mingzhe Li, Lixuan Wei, Zhenbang Du, **Xiangchi Yuan**, Dachuan Shi, Qirui Jin, Wenke Lee

Sparsity-Controllable Dynamic Top-p MoE for Large Foundation Model Pre-training [PDF]
International Conference on Machine Learning (ICML), 2026
 Can Jin, Hongwu Peng, Mingcan Xiang, Qixin Zhang, **Xiangchi Yuan**, Amit Hasan, Ohi Dibua,
 Yifan Gong, Yan Kang, Dimitris N. Metaxas

What Makes a Good Curriculum? Disentangling the Effects of Data Ordering on LLM Mathematical Reasoning [PDF] [Code]
The 64th Annual Meeting of the Association for Computational Linguistics (ACL Main), 2026
 Yaning Jia, Chunhui Zhang, Xingjian Diao, **Xiangchi Yuan**, Zhongyu Ouyang, Chiyu Ma, Soroush Vosoughi

MTMCS-Bench: Evaluating Contextual Safety of Multimodal Large Language Models in Multi-Turn Dialogues [PDF] [Code]
Findings of the Association for Computational Linguistics (ACL Findings), 2026
 Zheyuan Liu, Dongwhi Kim, Yixin Wan, **Xiangchi Yuan**, Zhaoxuan Tan, Fengran Mo, Meng Jiang

Modality-Aware Neuron Pruning for Unlearning in Multimodal Large Language Models [PDF] [Code]
The 63rd Annual Meeting of the Association for Computational Linguistics (ACL Main), 2025
 Zheyuan Liu, Guangyao Dou, **Xiangchi Yuan**, Chunhui Zhang, Zhaoxuan Tan, Meng Jiang

Growing Through Experience: Scaling Episodic Grounding in Language Models [PDF] [Code]
The 63rd Annual Meeting of the Association for Computational Linguistics (ACL Oral), 2025
 Chunhui Zhang, Sirui Wang, Zhongyu Ouyang, **Xiangchi Yuan**, Soroush Vosoughi

LongMamba: Enhancing Mamba’s Long-Context Capabilities via Training-Free Receptive Field Enlargement [PDF] [Code]
International Conference on Learning Representations (ICLR), 2025
 Zhifan Ye, Kejing Xia, Yonggan Fu, Xin Dong, Jihoon Hong, **Xiangchi Yuan**, Shizhe Diao, Jan Kautz,
 Pavlo Molchanov, Yingyan Celine Lin

AmoebaLLM: Constructing Any-Shape Large Language Models for Efficient and Instant Deployment [PDF] [Code]
Advances in Neural Information Processing Systems (NeurIPS), 2024
 Yonggan Fu, Zhongzhi Yu, Junwei Li, Jiayi Qian, Yongan Zhang, **Xiangchi Yuan**, Dachuan Shi,
 Roman Yakunin, Yingyan Celine Lin

HONORS AND AWARDS	Brandeis University Merit Scholarship (\$40,000)	2022 & 2023
	Brandeis University Conference Award	2023

TEACHING EXPERIENCE	Georgia Institute of Technology , Atlanta, GA	
	Teaching Assistant, CS6476 <i>Computer Vision</i>	Spring and Fall 2025
	Teaching Assistant, CS6262 <i>Network Security</i>	Spring and Fall 2026

	Brandeis University , Waltham, MA	
	Head Teaching Assistant, MATH-125A <i>Mathematics for Machine Learning</i>	Fall 2023

INVITED TALKS	Brandeis University Research Retreat	July 2023 & 2024
	Georgia Tech OMSCS Brown Bag Talks	March 2025

SERVICE	Program Committee/Conference Reviewer	
	ICML 2023-2026, ICLR 2024-2025, NeurIPS 2024-2025, EMNLP 2025-2026, ACL 2025-2026, UAI	